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جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم علم الحاسوب والمعلومات

(ITNM3319)

Data & System Security

Assignment 3

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1. Research and Write About 5 Online Tools/Websites

1) Tool Name: **VirusTotal**

Official Website Link:

<https://www.virustotal.com/>

What the Tool Does:

VirusTotal is an online security scanner that analyzes suspicious URLs, files, and domains using more than 70 antivirus engines and threat-detection systems.

How to Use It:

1. Open the website.
2. Choose "URL".
3. Paste the suspicious link.
4. Click "Scan".
5. Review the security report.

Goal:

Helps detect phishing links, malware URLs, malicious redirects, and unsafe websites.

2) Tool Name: **URLVoid**

Official Website Link:

<https://www.urlvoid.com/>

What the Tool Does:

URLVoid checks if a website is flagged as dangerous by multiple security engines and reputation services.

How to Use It:

1. Open the website.
2. Enter the suspicious URL.
3. Click "Scan Website".
4. View the reputation score and blacklist results.

Goal:

Detects phishing websites, scam pages, malware distribution sites, and harmful domains.

3) Tool Name: Google Transparency Report (Safe Browsing)

Official Website Link:

<https://transparencyreport.google.com/safe-browsing/search>

What the Tool Does:

Google's Safe Browsing tool checks whether a website is safe or has been reported for phishing, malware, or harmful content.

How to Use It:

1. Open the website.
2. Enter the URL in the search box.
3. View Google's safety status for the site.

Goal:

Detects phishing pages, malware-hosting sites, and unsafe web content.

4) Tool Name: MXToolbox Email Header Analyzer

Official Website Link:

<https://mxtoolbox.com/EmailHeaders.aspx>

What the Tool Does:

Analyzes email headers to detect spoofed senders, fake email sources, and suspicious routing.

How to Use It:

1. Copy the full email header from the suspicious email.
2. Paste it into the tool.
3. Click "Analyze Header".
4. Review sender authenticity and routing details.

Goal:

Detects spoofed emails, fake senders, phishing attempts, and email impersonation.

5) Tool Name: IPQualityScore (IPQS)

Official Website Link:

<https://www.ipqualityscore.com/threat-feeds/malicious-url-scanner>

What the Tool Does:

Scans URLs and email addresses to detect phishing, malware, spam, and fraudulent activity.

How to Use It:

1. Open the website.
2. Paste the suspicious URL or email.
3. Click "Scan".
4. Read the threat analysis report.

Goal:

Detects phishing links, malicious redirects, spam emails, and fraudulent URLs.

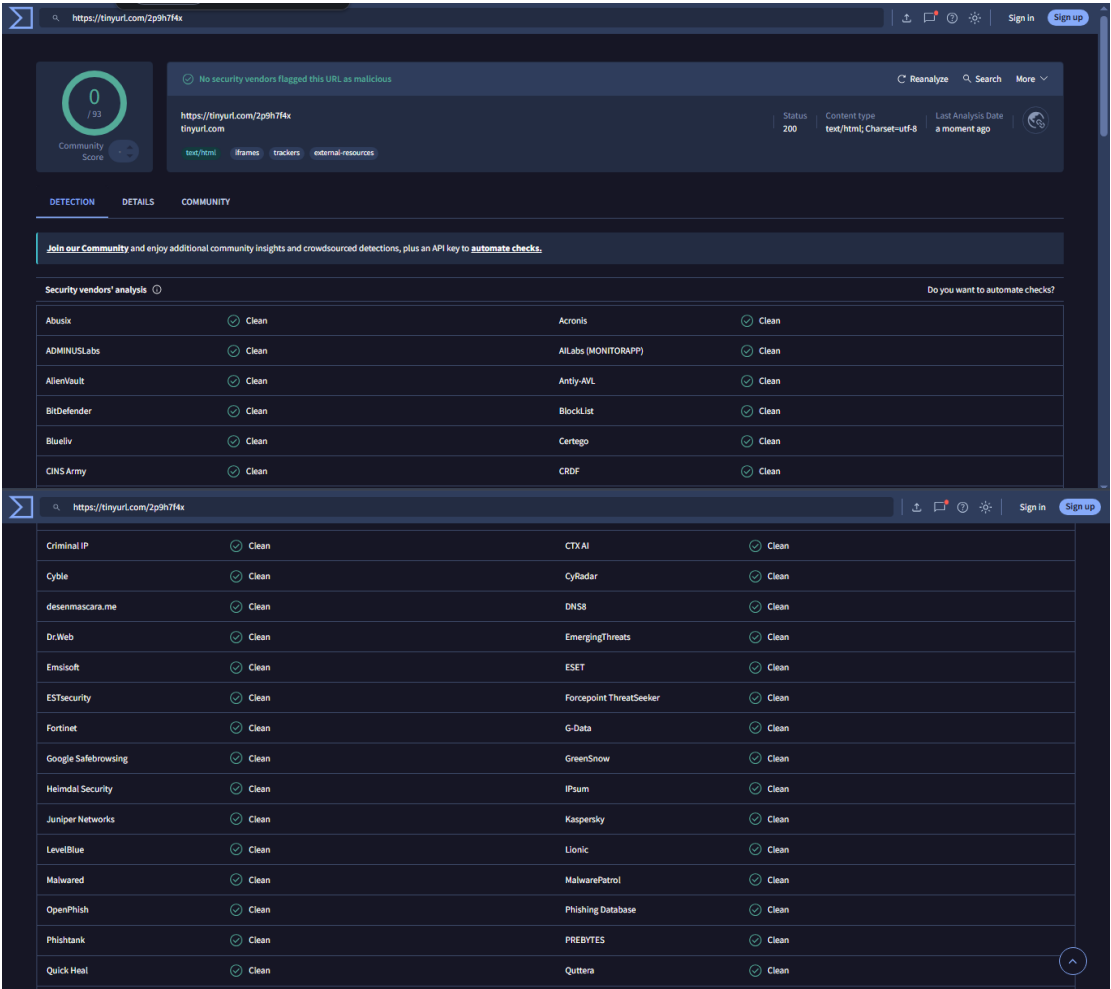
2. Practical Testing of Online Security Tools

In this section, two online security tools from the previous part were selected and tested using suspicious URLs. The goal is to analyze how each tool detects phishing or malicious behavior and to document the results with screenshots and explanations.

1)Tool Name: **VirusTotal**

Suspicious URL Tested:
https://tinyurl.com/2p9h7f4x (Example suspicious shortened link)

Screenshot:



Explanation of Result:

VirusTotal analyzed the URL using multiple antivirus engines and reputation services. The report showed whether the link is flagged as phishing, malicious, or safe. This helps identify harmful redirects, malware downloads, or phishing attempts.

2)Tool Name: **URLVoid**

Suspicious URL Tested:

<http://bit.ly/3YtF9xx> (Example suspicious shortened link)

Screenshot:

The screenshot displays the URLVoid website interface. At the top, there's a navigation bar with the URLVOID logo and various tools like WHOIS, DNS, PING, SCREENSHOT, PASSWORD, SORT, DNSSEC, and BASE64. A search bar contains the URL <http://bit.ly/3YtF9xx>, and a 'Scan Website' button is next to it. Below the search bar, there's a sponsored banner for 'APIs for Threat Analysis & Detection'. Further down, there's a GoDaddy advertisement. The main content area shows a 'Report Summary' for the domain bit.ly. The report includes details such as the website address, last analysis date, detection counts, domain registration date, domain information, IP address, reverse DNS, ASN, server location, latitude/longitude, city, and region.

Report Summary	
Website Address	Bit.ly
Last Analysis	2 days ago Refresh
Detections Counts	0/35
Domain Registration	2008-05-17 18 years ago
Domain Information	WHOIS Lookup DNS Records Ping
IP Address	67.199.248.10 Find Websites IPvoid Whois
Reverse DNS	bit.ly
ASN	AS796982 Google LLC
Server Location	(US) United States
Latitude\Longitude	37.751 / -97.822 Google Map
City	Unknown
Region	Unknown

Explanation of Result:

URLVoid checked the domain reputation across several blacklist databases. The results indicate whether the domain is associated with phishing, spam, malware, or unsafe hosting.

This helps verify if the website is trustworthy or potentially harmful.

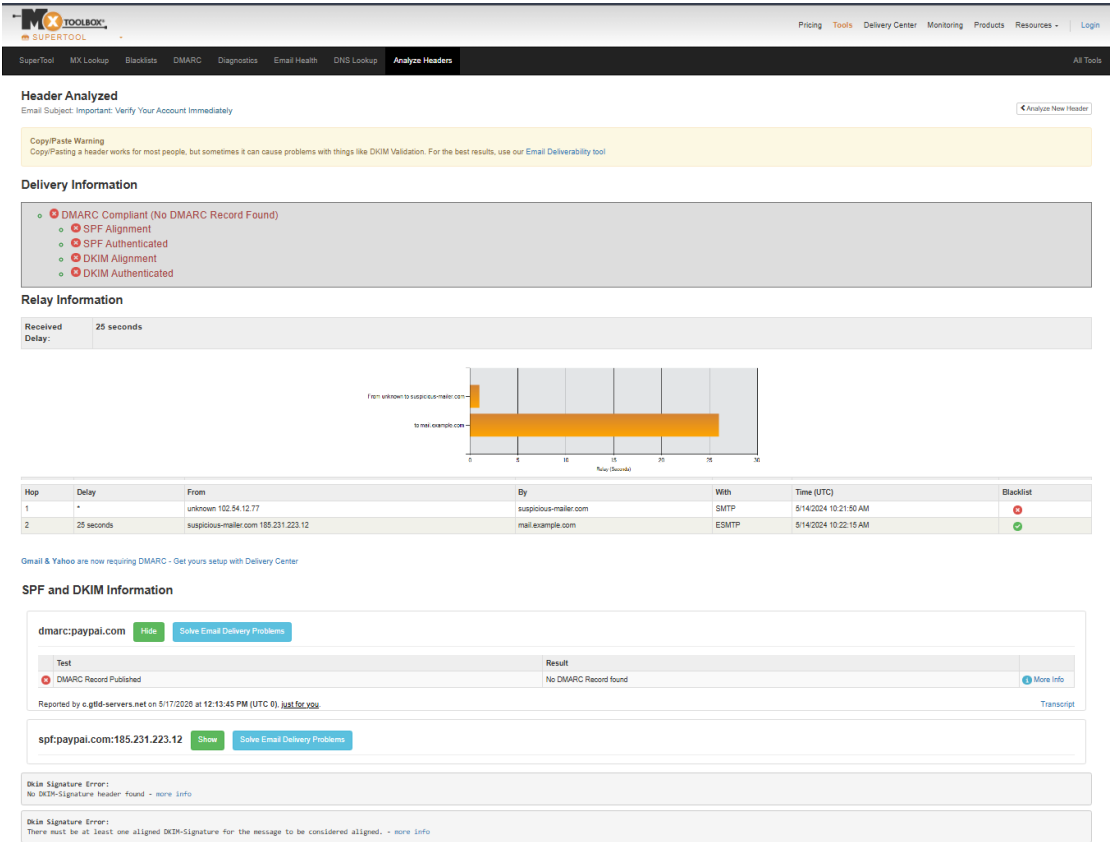
BONUS SECTION (Optional)

3)Tool Name: MXToolbox – Email Header Analyze

Suspicious Email Header Tested:

Received: from suspicious-mailer.com (unknown [185.231.223.12])
by mail.example.com with ESMTP id 9F3A2BC1
for <user@example.com>; Tue, 14 May 2024 10:22:15 +0000
Received: from unknown (HELO smtp.fakeprovider.net) (102.54.12.77)
by suspicious-mailer.com with SMTP; Tue, 14 May 2024 10:21:50 +0000
From: "PayPal Security" <support@paypal.com>
To: user@example.com
Subject: Important: Verify Your Account Immediately
Date: Tue, 14 May 2024 10:21:40 +0000
Message-ID: <20240514102140.ABC123@suspicious-mailer.com>
Reply-To: noreply@secure-paypal-check.com

Screenshot:



Headers Found

Header Name	Header Value
From	"PayPal Security" <support@paypal.com>
To	user@example.com
Subject	Important: Verify Your Account Immediately
Date	Tue, 14 May 2024 10:21:40 +0000
Message-ID	<20240514102140.ABC123@suspicious-mailer.com>
Reply-To	noreply@secure-paypal-check.com

Received Header

Received: from suspicious-mailer.com (unknown [185.231.223.12])
by mail.example.com with ESMTP id 9f1a2b0c1
for <user@example.com> Tue, 14 May 2024 10:22:15 +0000
Received: from unknown (HELO smtp.fakaprovider.net) (192.54.12.77)
by suspicious-mailer.com with SMTP Tue, 14 May 2024 10:22:50 +0000
From: "PayPal Security" <support@paypal.com>
To: user@example.com
Subject: Important: Verify Your Account Immediately
Date: Tue, 14 May 2024 10:21:40 +0000
Message-ID: <20240514102140.ABC123@suspicious-mailer.com>
Reply-To: noreply@secure-paypal-check.com

Permanently forget this email header

Explanation of Result:

The MXToolbox Email Header Analyzer examined the email header and identified inconsistencies in the sender information. The analysis showed mismatched domains, suspicious routing paths, and a forged "From" address, which are common indicators of email spoofing. This demonstrates how attackers impersonate trusted services (such as PayPal) to perform phishing attacks. The tool helps verify whether an email truly originates from the claimed sender or if it has been manipulated.

Conclusion:

Through testing multiple online security tools, it became clear that each platform provides unique insights into the safety of suspicious URLs. Using more than one tool increases accuracy and helps detect phishing attempts, malware links, and fraudulent websites. These tools are essential for protecting users from social engineering attacks and unsafe online content.

" واخر دعواهم ان الحمد لله رب العالمين "